# Eliza Hahn Neights

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## Personal Statement

I am a Physics PhD student at George Washington University studying gamma-ray bursts (GRBs) at NASA Goddard Space Flight Center. I make predictions for next-generation GRB science that will be done with the upcoming space telescope COSI, write the spectral and polarization analysis software for COSI, and am the Early Career Co-Lead for the COSI GRB science team. I also study GRBs observed by *Fermi*-GBM, for which I serve as a Burst Advocate.

## Education

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|---|---|---|
| 2021-2027 | **George Washington University** | Ph.D in Physics (*in progress*)<br>*Advisors: Dr. Sylvain Guiriec & Dr. Carolyn Kierans*<br>**Thesis:** Studying Gamma-Ray Bursts with COSI and Fermi-GBM Analysis of a Record Ultra-long Gamma-Ray Burst |
| 2021-2024 | **George Washington University** | M.Phil in Physics<br>M.S. in Physics |
| 2017-2021 | **Pennsylvania State University** | B.S. in Astronomy & Astrophysics<br>B.S. in Physics<br>Minor in Mathematics |

## Leadership Roles

2026-present **Early Career Co-Lead for COSI GRB Science Team**
First Early Career Co-Lead for a COSI science team and first student member of COSI Central Science Team. Manage >50 person group and organize monthly meetings.

2024-present **Burst Advocate for Fermi-GBM**
Volunteer for 6-8 shifts per month. Analyze triggered observations, ensure good data quality, and send alerts to scientific community.

2022-present **Spectral and Polarization Analysis Software Development for COSI**
Lead development of publicly available spectral and polarization analysis software used by scientific community to analyze COSI data. github.com/cositools/cosipy

2023-2026 **Technical Lead for COSI GRB Science Team**
Led creation of simulated COSI transient observations used to develop analysis software, test onboard trigger algorithm, and make GRB science predictions. Developed software to simulate realistic population-level COSI observations of transients. Managed GRB team tasks, including development of COSI transient analysis pipelines. Planned and led monthly meetings. github.com/cositools/grb-simulations

## Awards & Scholarships

2026-2027 **Chateaubriand Fellowship**
Grant awarded by French Embassy's Office for Science and Technology to conduct GRB research using SVOM at Institute of Research in Astrophysics and Planetary Science in France for six months.

2025 **Gus W. Weiss Prize (Honorable Mention)**
Award celebrating graduate student research contributions and dedication to George Washington University Physics department and larger physics community.

## Awards & Scholarships (continued)

2022-2025 **John Mather Nobel Scholarship**
Travel allowance awarded by John and Jane Mather Foundation for Science and the Arts towards cost of presenting research at conferences.

2017-2021 **Braddock Scholarship**
Four-year academic merit scholarship awarded by Penn State Eberly College of Science.

2019-2021 **M. Dean and Jean L. Underwood Scholarship in Physics**
Academic merit scholarship awarded by Penn State Eberly College of Science.

2019-2020 **Eberly College of Science Undergraduate Research Award**
Funding from Office of Science Engagement in Penn State Eberly College of Science to do undergraduate astrophysics research with Dr. Doug Cowen.

2017-2020 **Dean's List**
Maintained GPA above 3.5.

2018 **NASA Pennsylvania Space Grant**
Women in Science and Engineering Research scholarship to do undergraduate astrophysics research with Dr. Miguel Mostafá.

## Publications

In addition to the publications below, I have led 14 GCN circulars and 1 Astronomer's Telegram.
E. Neights et al. 2026 was featured as a *Nature Astronomy* Research Highlight in Feb 2026.
Orcid: `0009-0005-0762-4507`, SciX

### Refereed


**E. Neights**, E. Burns, C. L. Fryer, *et al.*, "GRB 250702B: discovery of a gamma-ray burst from a black hole falling into a star," *MNRAS*, Jan. 2026. DOI: `10.1093/mnras/staf2019`.

N. Parmiggiani, A. Bulgarelli, G. Panebianco, *et al.*, "COSI Short Gamma-Ray Burst Localization Using BGO Shield Data," *ApJ*, Feb. 2026. DOI: `10.3847/1538-4357/ae25fc`.

A. Rizzo, N. Parmiggiani, A. Bulgarelli, *et al.*, "Comparing Classical and Quantum Deep Learning Techniques for Anomaly Detection of Short-Duration Gamma-Ray Signals," *Astronomy and Computing*, Jul. 2026. DOI: `10.1016/j.ascom.2026.101090`.

A. C. Trigg, E. Burns, M. Negro, *et al.*, "From Rare Events to a Population: Discovering Overlooked Extragalactic Magnetar Giant Flare Candidates in Archival Fermi Gamma-ray Burst Monitor Data," *arXiv e-prints*, Oct. 2025. DOI: `10.48550/arXiv.2510.23367`.

H. Yoneda, T. Siegert, I. Martinez-Castellanos, *et al.*, "Enhancing Compton telescope imaging with maximum a posteriori estimation: A modified Richardson–Lucy algorithm for the Compton Spectrometer and Imager," *A&A*, May 2025. DOI: `10.1051/0004-6361/202453528`.

H. C. Gulick, **E. Neights**, S. Al Nussirat, *et al.*, "Across the soft gamma-ray regime: utilizing simultaneous detections in the Compton Spectrometer and Imager (COSI) and the Background and Transient Observer (BTO) to understand astrophysical transients," in *Space Telescopes and Instrumentation 2024: Ultraviolet to Gamma Ray*, Aug. 2024. DOI: `10.1117/12.3020606`.


### Non-Refereed


N. Parmiggiani, A. Bulgarelli, G. Panebianco, *et al.*, "Localization and Confidence Region Estimation of Short GRBs with the COSI BGO Shield Using a HEALPix-Based Deep Learning Approach," *arXiv e-prints*, Apr. 2026. DOI: `10.48550/arXiv.2604.15119`.

E. Burns, J. Andrews, R. Szabo, *et al.*, "The Heavy Element Enrichment History of the Universe from Neutron Star Mergers with Habitable Worlds Observatory," *arXiv e-prints*, Jul. 2025. DOI: `10.48550/arXiv.2507.09778`.

E. Burns, C. L. Fryer, I. Agullo, *et al.*, “Multidisciplinary Science in the Multimessenger Era,” *arXiv e-prints*, Feb. 2025. DOI: `10.48550/arXiv.2502.03577`.

H. Yoneda, T. Siegert, I. Martinez-Castellanos, *et al.*, “Imaging MeV Gamma-ray Lines with Advanced Image Reconstruction Framework for COSI,” in *39th International Cosmic Ray Conference*, Dec. 2025. URL: `https://pos.sissa.it/501/891/pdf`.

J. Tomsick, S. Boggs, A. Zoglauer, *et al.*, “The Compton Spectrometer and Imager,” in *38th International Cosmic Ray Conference*, Sep. 2024. DOI: `10.22323/1.444.0745`.

I. Martinez-Castellanos, S. Gallego, C.-Y. Huang, *et al.*, “The cosipy library: COSI’s high-level analysis software,” *arXiv e-prints*, Aug. 2023. DOI: `10.48550/arXiv.2308.11436`.

T. Akindele, T. Anderson, E. Anderssen, *et al.*, “A Call to Arms Control: Synergies between Nonproliferation Applications of Neutrino Detectors and Large-Scale Fundamental Neutrino Physics Experiments,” *arXiv e-prints*, Feb. 2022. DOI: `10.48550/arXiv.2203.00042`.

T. Grégoire, H. Ayala Solares, S. Coutu, *et al.*, “Model independent search for transient multimessenger events with AMON using outlier detection methods,” in *37th International Cosmic Ray Conference*, Mar. 2022, p. 934. DOI: `10.22323/1.395.0934`.

## Presentations

### Invited Talks

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|---|---|
| Jun 2026 | **COSI Observations of GRBs** “Exploring the MeV Gamma-ray Sky with COSI” Special Session at the American Astronomical Society Meeting (*Pasadena, CA*) |
| Mar 2026 | **Discovery of a Gamma-Ray Burst from a Black Hole Falling into a Star** American Physical Society Global Physics Summit (*Denver, CO*) |
| Oct 2025 | **COSI GRB Group Update** COSI Collaboration Meeting (*St. Louis, MO*) |
| Oct 2024 | **COSI’s Spectral Analysis Software** COSI Collaboration Meeting (*Berkeley, CA*) |
| Oct 2024 | **GRB Polarization with COSI** COSI Collaboration Meeting (*Berkeley, CA*) |
| Apr 2024 | **The COSI GRB Science Goals** “Exploring the MeV Gamma-ray Sky: The Past, Present, and Future” Special Session at the High Energy Astrophysics Division Meeting (*Horseshoe Bay, TX*) |
| Jul 2024 | **GRB Polarization with COSI** GRB Forum (*Athens, Greece*) |

### Contributed Talks

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|---|---|
| Jun 2026 | **Gamma-Ray Burst from a Black Hole Falling into a Star** American Astronomical Society Meeting (*Pasadena, CA*) |
| | **Probing Gamma-Ray Bursts with Observations of Ultralong GRBs and the Upcoming COSI Mission** Seminar at California Institute of Technology (*Pasadena, CA*) |
| May 2026 | **Studying GRBs with COSI** Seminar at the Flatiron Institute’s Center for Computational Astrophysics (*New York, NY*) |
| | **Gamma-Ray Burst from a Black Hole Falling into a Star** Theoretical High Energy Astrophysics Seminar at Columbia University (*New York, NY*) |
| Oct 2025 | **GRB Localizations and Polarimetry with COSI** “Exploring the MeV Gamma-ray Sky with COSI” Special Session at the High Energy Astrophysics Division Meeting (*St. Louis, MO*) |
| Jan 2025 | **Gamma-Ray Burst Polarization with COSI** American Astronomical Society Meeting (*National Harbor, MD*) |
| Aug 2024 | **Studying GRBs Using COSI’s Polarization Measurements** AstroCon DC (*Washington, DC*) |
| Jun 2024 | **Studying GRBs Using COSI** *Fermi* Summer School (*Lewes, DE*) |
| May 2024 | **GRB Science Using COSI** Seminar at Oregon State University (*Corvallis, OR*) |

## Presentations (continued)

Apr 2024 **Studying Gamma-Ray Bursts Using COSI** High Energy Astrophysics Division Meeting (*Horseshoe Bay, TX*)

Aug 2023 **Using COSI to Study GRBs** DMV Consortium Graduate Student Astrophysics Conference (*Washington, DC*)

Aug 2022 **Scientific Pipeline Development for COSI** NASA Goddard Space Flight Center Astrophysics Summer Intern Symposium (*Virtual*)

Sep 2020 **Archival Coincidence Analysis for IceCube Cascade Events & GRBs** IceCube Collaboration Meeting (*Virtual*)

### Posters

Oct 2025 **Gamma-Ray Observations of GRB 250702B: Gamma-Ray Burst from a Black Hole Falling into a Star** High Energy Astrophysics Division Meeting (*St. Louis, MO*)

Mar 2023 **Introduction to COSI Data Analysis** High Energy Astrophysics Division Meeting (*Waikoloa, HI*)

May 2020 **Photomultiplier Tube Pressure Testing for WATCHMAN** Pennsylvania State University Undergraduate Exhibition (*State College, PA*)

Nov 2018 **The Study of Binary Neutron Star Mergers from Simulations** Pennsylvania State University Undergraduate Research Symposium (*State College, PA*)

## Selected Press

2026 **BBC Sky at Night**
`skyatnightmagazine.com/news/grb-250702b-study-interview`

2025 **National Geographic**
`nationalgeographic.com/science/article/gamma-ray-burst-black-hole-eats-star`

**NASA**
`science.nasa.gov/science-research/black-hole-eats-star`

**Sky and Telescope**
`skyandtelescope.org/astronomy-news/black-hole-eats-through-star-explodes-it-from-within`

**GW Today**
`gwtoday.gwu.edu/black-hole-eats-star-student-charts-record-blast`

## Teaching

2022 **Teaching Assistant for Classical Mechanics**
Assisted with lesson delivery, graded, held weekly office hours, and led one independent lecture for an undergraduate electricity and magnetism class of 38 students.

2021 **Teaching Assistant for Electricity & Magnetism**
Assisted with lesson delivery, graded, and held weekly office hours for an undergraduate electricity and magnetism class of 25 students.

## Outreach & Professional Development

2025 **Attendee of Summer School in Statistics for Astronomers**

2024 **Attendee of Fermi Summer School**

**Volunteer for Astronomy Festival on the National Mall**

## Outreach & Professional Development (continued)

2023-2024 **Member of COSI Collaborations Connections Team** Addressed issues of diversity, equity, inclusion, and accessibility in COSI collaboration.

2020 **Volunteer for Penn State AstroFest**

2019-2021 **Safety Liaison for Penn State Social Dance Club**
Wrote safety and inclusivity policies, addressed concerns of members, and handled cases of misconduct.

2018 **Presenter at Penn State AstroFest**

## Collaborations

**COSI**
Early Career Co-Lead for GRB science team, lead creation of transient observation simulations, make GRB science predictions, and develop spectral and polarization analysis software.

**Fermi-GBM**
Burst Advocate and led analysis of record-duration GRB 250702B.